

Performance Evaluation of Detection and Tracking Algorithms for Automated Grape Cluster Counting

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Abstract—The accurate estimation of crop yield is crucial in modern agriculture; however, traditional methods, such as manual counting of grape clusters, often result in errors exceeding 20% and are impractical for large-scale monitoring under variable environmental and climatic conditions. While object detection techniques have been successfully implemented in real-world agricultural scenarios, the integration and performance of multi-object tracking (MOT) algorithms in such dynamic and unstructured environments remains underexplored. This article provides a comparative evaluation of state-of-the-art object detection and tracking methodologies for use in vineyard conditions. Here, two detection methods—YOLOv12 and RF-DETR—were compared in combination with two tracking algorithms—BoT-SORT and ByteTrack—to generate four distinct configurations aimed at the automatic counting of grape clusters. Experimental results on a video dataset recorded in agroindustrial vineyards showed that the combination of YOLOv12 with BoT-SORT achieved the highest average relative counting accuracy, with 84.31%. These findings confirm the potential of AI-based techniques for precision agriculture, offering greater accuracy in yield estimation, optimized resource management, and a substantial reduction in reliance on manual labor.

Keywords—Grape, Object Detection, Object Tracking, Precision Agriculture

I. INTRODUCTION

In modern agriculture, efficient management and monitoring of agricultural resources are fundamental for optimizing performance and ensuring sustainable practices. In viticulture, accurate grape cluster counting directly influences production prediction, harvest scheduling, and resource allocation. However, conventional fruit counting methods depend on human experience and availability, with error rates exceeding 20% under adverse conditions, generating production cost overruns [1].

Advances in computer vision, particularly deep learning, have improved agricultural automation by overcoming manual counting limitations. YOLO architectures have demonstrated 92% accuracy in grape cluster identification under controlled conditions using RGB images [2].

While object detection provides accurate identification of grape clusters in individual frames, counting across time or video sequences requires tracking each instance to avoid duplicates or omissions. Therefore, to achieve reliable counting in dynamic agricultural environments, it is essential to integrate object detection with multi-object tracking (MOT) algorithms such as BoT-SORT and ByteTrack, which address these issues by maintaining object identities across frames, enabling more accurate counting even in dense vineyard canopies [3].

Integrated real-time detection and tracking approaches show potential for automating grape cluster counting. AgriSORT implements the *tracking-by-detection* paradigm achieving 66.13% MOTA in real vineyard environments [4]. In [5], YOLOv3 integrated with depth sensors achieves 95.7% accuracy and 22 FPS in simulated environments using spatial positioning logic for cluster detection. In [6], a hybrid Swin-Transformer-YOLOv5 architecture obtains 97.4% mAP in luminance variability scenarios, but without tracking modules, limiting utility in dynamic tasks.

Despite these advances, key limitations persist. System performance decreases under abrupt lighting changes, severe occlusion, or dense overlap. Studies such as [7] show 15% detection accuracy drops in poor lighting, highlighting the need for robust models in unstructured outdoor conditions. Furthermore, systematic comparative analyses of modern detector-tracker combinations under these challenging conditions remain underexplored.

This work presents a comparison of YOLOv12 and RF-DETR architectures integrated with BoT-SORT and ByteTrack algorithms for grape cluster counting. Building on established individual algorithm performance, this research examines how modern detection-tracking combinations perform under adverse field conditions including partial occlusions, motion blur, and high object density. The results offer empirical evidence on the behavior of these integrated detector-tracker systems in challenging agricultural environments.

II. MATERIALS AND METHODS

A. Data Acquisition and Detection Dataset

Data were collected in a vineyard during the final maturation stage of the Allison grape variety. Images and videos were captured under natural conditions to ensure the relevance of the study in real agroindustrial scenarios. A total of 880 images with a resolution of 4080×3072 pixels were collected, which served as the basis for model training and validation.

The dataset was split into 80% for training (704 images), 10% for validation (88 images), and 10% for testing (88 images). Bounding boxes for grape clusters were generated semi-automatically using the GroundingDINO model [8] and manually verified to ensure labeling accuracy. To enhance model robustness and prevent overfitting, data augmentation techniques—such as random horizontal flips and color adjustments—were applied exclusively to the training set. As a result, the training set was expanded to 2112 images.

The complete process for generating the grape cluster detection dataset is illustrated in Figure 1.

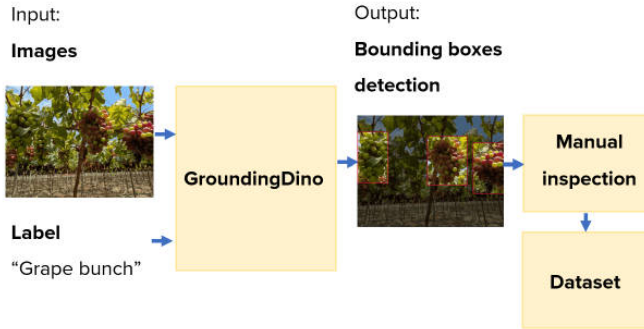


Fig. 1: Dataset generation pipeline for grape cluster detection.

B. Detection Algorithms Configuration

The selection of YOLOv12 and RF-DETR was based on their complementary characteristics. YOLOv12 is recognized for its high inference speed and ease of implementation, making it suitable for real-time applications on devices with limited computational resources. On the other hand, RF-DETR employs a transformer-based architecture that is robust in complex environments. This combination allowed for a comparative analysis between convolutional neural network (CNN)-based and transformer-based object detection models under standardized conditions.

1) *YOLOv12*: Introduces an attention-centered architecture that departs from the traditional CNN-based approaches used in previous YOLO models, while preserving the real-time inference speed essential for many applications. This model achieves high accuracy in object detection through novel methodological innovations in attention mechanisms and overall network architecture, while maintaining real-time performance [9].

In this study, the YOLOv12l (large) variant was employed with a total of 60 epochs for training, a batch size of 16, the SGD optimizer with momentum, and a learning rate of 10^{-4} .

The loss function used was binary cross-entropy (BCE) for classification and objectness tasks, and Complete IoU (CIoU) loss for bounding box regression. The respective configuration can be observed in Table I.

TABLE I
YOLOv12L TRAINING HYPERPARAMETERS

Parameter	Value
Input Size	640×640
Batch Size	16
Optimizer	SGD with momentum
Learning Rate	10^{-4}
Epochs	60
Loss Function	BCE (cls/obj), CIoU (bbox)
Model Variant	YOLOv12l

2) *RF-DETR*: It is based on the DETR (DEtection TRansformer) architecture [10], which replaces traditional region proposal networks with a transformer-based encoder-decoder structure. This approach eliminates anchor boxes and post-processing stages such as non-maximum suppression (NMS). Through multi-head attention mechanisms, RF-DETR captures long-range dependencies and models global context, making it advantageous for scenarios with occlusions or overlapping objects. The transformer decoder enables direct set prediction and end-to-end training for consistent object localization [11].

Training was performed with 640×640 pixel input resolution using the AdamW optimizer with a learning rate of 10^{-4} for 60 epochs. The loss function combined cross-entropy for classification with L1 and Generalized IoU (GIoU) losses for bounding box regression. Training hyperparameters are summarized in Table II.

TABLE II
RF-DETR TRAINING HYPERPARAMETERS

Parameter	Value
Input Size	640×640
Batch Size	16
Optimizer	AdamW
Learning Rate	10^{-4}
Epochs	60
Loss Function	Cross-Entropy, L1, GIoU
Model Variant	RF-DETR Base

C. Tracking Algorithms Configuration

To ensure temporal consistency in object trajectories, the detection outputs from YOLOv12 and RF-DETR are coupled with two multi-object tracking (MOT) algorithms: ByteTrack and BoT-SORT. These trackers were selected based on their proven effectiveness in handling challenges such as occlusions, variable motion, and identity preservation in video sequences. A high-level overview of the input–output relationship in the detection and tracking process is shown in Figure 2.

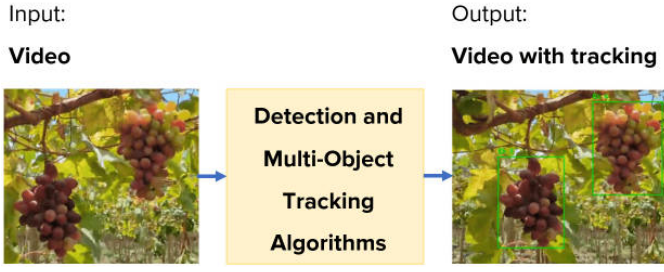


Fig. 2: High-level overview of the object tracking process.

1) *ByteTrack*: It introduces a dual-threshold association mechanism that improves object recovery under occlusion by incorporating low-confidence detections during the data association process [12]. It uses high-confidence detections for primary matching and low-confidence detections (between 0.1 and 0.5) to recover targets that have been temporarily occluded. Additionally, it implements a Kalman filter for motion prediction and employs a cost matrix based on Intersection over Union (IoU) to optimize trajectory assignment. Five distinct parameter configurations were evaluated for experimental validation, which are summarized in Table III.

TABLE III
BYTETRACK PARAMETER CONFIGURATIONS

Parameter	Conf. 1	Conf. 2	Conf. 3	Conf. 4	Conf. 5
Detection Threshold	0.85	0.80	0.78	0.75	0.82
Track Buffer (frames)	300	200	150	250	180
Matching Threshold	0.80	0.75	0.70	0.65	0.72

2) *BoT-SORT*: It is based on traditional tracking-by-detection frameworks through the integration of additional modules that enhance robustness in complex scenarios [13]. Specifically, it incorporates a re-identification (ReID) module based on appearance features to preserve object identity over time. It also integrates optical flow to compensate for camera movements and utilizes a Kalman filter with adaptive noise covariance for more accurate motion prediction. Data association is performed using the Hungarian algorithm, employing cosine distance between feature vectors as the similarity metric. The parameter sets evaluated for this algorithm are presented in Table IV.

TABLE IV
BoT-SORT PARAMETER CONFIGURATIONS

Parameter	Conf. 1	Conf. 2	Conf. 3	Conf. 4	Conf. 5
Detection Threshold	0.85	0.80	0.78	0.75	0.82
Track Buffer (frames)	300	200	150	250	180
Matching Threshold	0.80	0.75	0.70	0.65	0.72

D. Experimental Setup

The performance of two object detection models, YOLOv12 and RF-DETR, was evaluated in combination with two multi-object tracking algorithms, ByteTrack and BoT-SORT. While the detection models had no additional changes after their training, the tracking algorithms were subjected to five distinct parameter configurations, as detailed in Tables III and IV. Each parameter set was applied to 10 video sequences, resulting in

the experimental combinations summarized in Table V. This structured overview provides the basis for the subsequent analysis, enabling a systematic benchmark comprising 200 total runs and facilitating a comprehensive evaluation of detection and tracking performance under variable field conditions.

TABLE V
EXPERIMENTAL COMBINATIONS AND TOTAL RUNS

Detector	Tracker	Param. Conf.	Videos	Total Runs
YOLOv12	ByteTrack	5 sets	10	50
	BoT-SORT	5 sets	10	50
RF-DETR	ByteTrack	5 sets	10	50
	BoT-SORT	5 sets	10	50

The ground truth value for grape clusters was obtained through visual inspection, with each video manually reviewed by a human observer to determine the total number of clusters. The video sequences were selected to represent real-world challenges such as occlusion and motion blur, examples of these conditions are illustrated in Figure 3, highlighting motion blur during acquisition and occlusions from foliage. The characteristics of these sequences are summarized in Table VI, which includes video duration, manual count, and the presence of specific challenging conditions.

TABLE VI
VIDEO DATASET CHARACTERISTICS AND MANUAL OBJECT COUNTS

Video ID	Duration (s)	Manual Count	Challenging Conditions
1	70	85	Motion blur
2	45	85	Occlusions
3	83	159	Occlusions
4	87	105	Motion blur
5	91	113	Motion blur
6	85	130	Occlusions
7	89	89	Motion blur
8	89	140	Occlusions
9	34	65	Occlusions
10	91	170	Motion blur



(a) Motion blur



(b) Occlusion

Fig. 3: Comparison of grape bunch visibility under different challenging conditions. Image (a) demonstrates motion blur effects that can occur during data acquisition, while image (b) presents occlusion due to overlapping leaves and grape clusters.

E. Evaluation Metrics

1) *Mean Average Precision*: The detection model performance was evaluated using mean average precision (mAP)

at different Intersection over Union (IoU) thresholds. mAP is calculated as the mean of average precision (AP) across all object classes:

$$\text{mAP@50} = \frac{1}{N} \sum_{i=1}^N \text{AP}_{0.5}^{(i)} \quad (1)$$

$$\text{mAP@50:95} = \frac{1}{10N} \sum_{t=0.5}^{0.95} \sum_{i=1}^N \text{AP}_t^{(i)} \quad (2)$$

where t ranges from 0.5 to 0.95 in steps of 0.05, and N is the number of object classes.

2) *Counting Accuracy Metrics*: To evaluate grape cluster counting accuracy, the number of unique object identities produced by the tracking algorithms was compared against manually obtained ground truth values through visual counting. The tracking-based count was calculated by aggregating unique IDs maintained throughout each video sequence, providing a consistent basis for evaluating object identity preservation under occlusion and lighting changes.

Three metrics were used to assess counting performance. Mean Absolute Error (MAE) measures the average absolute difference between predicted and actual counts, treating all errors equally:

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |C_{\text{pred}}^{(i)} - C_{\text{true}}^{(i)}| \quad (3)$$

where $C_{\text{pred}}^{(i)}$ is the predicted count for video i , $C_{\text{true}}^{(i)}$ is the ground truth count for video i , and n is the total number of videos.

Mean Relative Error (MRE) expresses counting error as a percentage of the true count, enabling comparison across videos with different cluster densities:

$$\text{MRE} = \frac{1}{n} \sum_{i=1}^n \frac{|C_{\text{pred}}^{(i)} - C_{\text{true}}^{(i)}|}{C_{\text{true}}^{(i)}} \times 100\% \quad (4)$$

Root Mean Square Error (RMSE) penalizes larger counting errors more heavily by squaring differences before averaging, emphasizing the importance of avoiding significant miscounts:

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (C_{\text{pred}}^{(i)} - C_{\text{true}}^{(i)})^2} \quad (5)$$

Lower values of all three metrics indicate higher counting accuracy.

III. RESULTS

A. Training Performance Comparison

Figures 4 and 5 show the performance of the YOLOv12 and RF-DETR models, respectively, in terms of mean Average Precision at IoU thresholds from 0.50 to 0.95 (mAP@[.50:.95]) across 60 epochs. These curves highlight the convergence behavior and learning dynamics of each detection architecture.

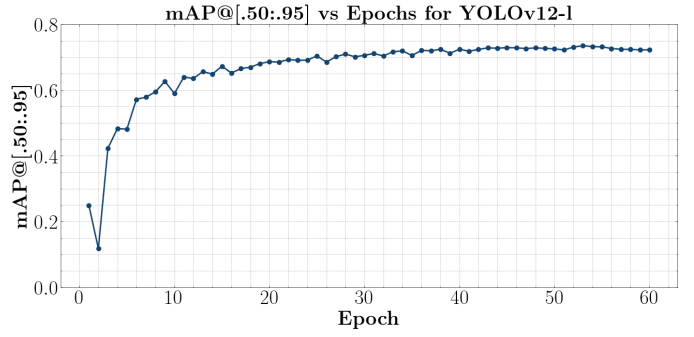


Fig. 4: YOLOv12 detection performance across 60 epochs, measured by mAP@[.50:.95]. The model gradually improves, reaching 0.73 at convergence.

YOLOv12 demonstrates a steady performance trajectory, starting with relatively low mAP values near 0.25 and steadily improving throughout the first 30 epochs. After epoch 35, mAP stabilizes with fluctuations between 0.70 and 0.73. This suggests effective learning and generalization as training progresses.

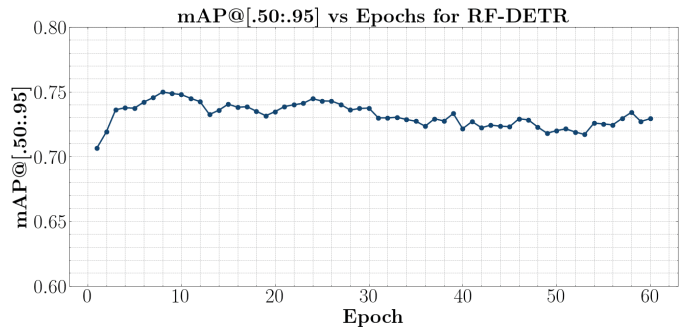


Fig. 5: RF-DETR detection performance across 60 epochs, measured by mAP@[.50:.95]. The model starts strong with high mAP values around 0.71 and remains stable throughout.

In contrast, RF-DETR starts with a high mAP value of 0.71 from the first epoch, showing minimal variation over time. The curve remains relatively flat, fluctuating slightly between 0.72 and 0.74. This suggests that RF-DETR benefits from strong pretraining or architectural advantages, enabling early convergence. However, the lack of further improvement may indicate limited responsiveness to continued optimization.

Overall, YOLOv12 exhibits a more typical learning curve with progressive gains, while RF-DETR delivers immediate but plateaued performance. These trends align with downstream results in object counting and tracking, where YOLOv12 showed greater adaptability across diverse scenarios.

B. Object Counting Results by Model-Tracker Combinations

Figures 6 to 9 present the object count results across ten videos for each of the five tested configurations per model-tracker combination. The manual count serves as the ground truth reference for evaluating algorithmic accuracy. These

visualizations, in conjunction with the aggregated metrics in Table VII, provide a comprehensive view of the performance patterns and disparities among the tested methods.

YOLOv12 with BoT-SORT stands out for its consistency and proximity to manual counts across all videos and configurations, as seen in Figure 6. The visual alignment of the bars with the ground truth highlights its low variability and strong reliability. This is further corroborated by its lowest Mean Absolute Error (MAE = 17.16) and Mean Relative Error (MRE = 15.69%), indicating robust object detection and effective track continuity. The low standard deviation (34.41) across configurations confirms stable performance.

Notably, parameter configuration 3 delivered the best individual results within this combination, achieving the lowest MAE of 12.10, MRE of 10.89%, and RMSE of 14.84. These values reflect superior counting accuracy, highlighting this configuration's optimal balance between detection sensitivity and tracking robustness.

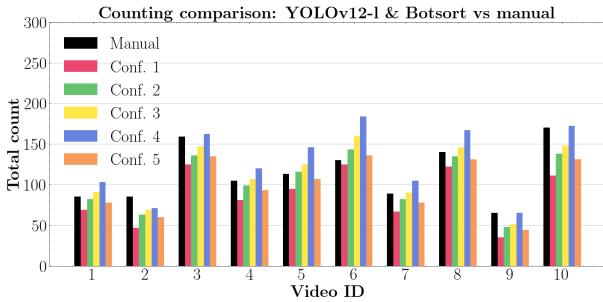


Fig. 6: YOLOv12 with BoT-SORT across five configurations. It shows high consistency with manual counts and lower absolute errors.

Replacing BoT-SORT with ByteTrack while retaining YOLOv12 as the detector yields slightly diminished but still competitive results (see Figure 7). Although the MAE (18.74) and MRE (17.03%) remain low, there is an observable increase in variability across configurations, as shown by the higher standard deviation (35.54). These findings suggest that ByteTrack, while capable of maintaining overall trends, may be more sensitive to parameter tuning and less robust in handling crowded or ambiguous tracking scenarios.

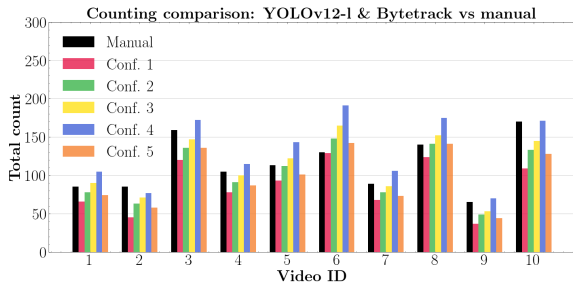


Fig. 7: YOLOv12 with ByteTrack. It maintains good correlation but shows slightly higher dispersion compared to BoT-SORT.

RF-DETR combined with BoT-SORT (Figure 8) shows a noticeable increase in counting errors, particularly in videos with high object density. Despite leveraging the same tracking algorithm as the YOLOv12-based approach, this combination recorded a higher MAE (23.40) and MRE (19.88%), pointing toward detection-related inaccuracies. The tendency to overcount likely stems from RF-DETR generating false positives, which are then propagated and reinforced by the tracker. Nevertheless, the general detection pattern remains aligned with the ground truth, suggesting that while counts are generally increased, the relative counting trends are preserved.

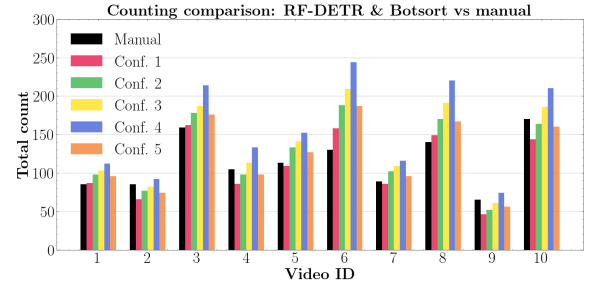


Fig. 8: RF-DETR with BoT-SORT. The results tend to overcount in higher-density scenes, leading to larger deviations.

The combination of RF-DETR and ByteTrack (Figure 9) produced the weakest performance overall, recording the highest errors across all metrics, with an MAE of 29.28 and an MRE of 22.12%. The elevated standard deviation (52.22) and considerable variability across configurations suggest limited consistency and adaptability. These outcomes likely stem from compounded issues, including overcounting at the detection stage and unstable tracking behavior due to frequent ID switches or fragmented trajectories.

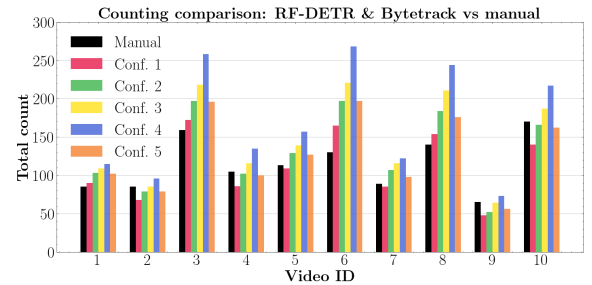


Fig. 9: RF-DETR with ByteTrack. It presents the highest absolute and relative errors among all combinations, indicating potential instability.

The aggregated results in Table VII support the visual trends observed in the figures. YOLOv12, particularly when combined with BoT-SORT, consistently outperforms the other configurations in both accuracy and stability. In contrast, RF-DETR's performance suffers from a combination of overcounting and increased tracking inconsistencies when paired with ByteTrack. The differences in MAE, MRE, and RMSE clearly differentiate the effectiveness of each approach. These results

highlight the critical impact of pairing accurate object detectors with reliable tracking algorithms, especially in dynamic or densely populated scenes.

TABLE VII
AVERAGE PERFORMANCE METRICS BY MODEL AND TRACKING ALGORITHM

Detector	Tracker	MAE	MRE (%)	RMSE	Std. Dev.
YOLOv12	BoT-SORT	17.16	15.69	20.80	34.41
	ByteTrack	18.74	17.03	22.62	35.54
RF-DETR	BoT-SORT	23.40	19.88	29.84	46.91
	ByteTrack	29.28	22.12	37.82	52.22

IV. CONCLUSIONS

This study evaluated object counting performance across ten videos using four detector-tracker combinations. YOLOv12 with BoT-SORT achieved the lowest MRE (15.69%), corresponding to the highest relative counting accuracy of 84.31%, along with the lowest MAE (17.16) and RMSE (20.80), making it the most consistent pairing overall. Among the five configurations tested, parameter configuration 3 yielded the best results, with an MAE of 12.10, MRE of 10.89%, and RMSE of 14.84, highlighting the importance of careful parameter tuning.

In contrast, RF-DETR exhibited a tendency to overestimate counts, with this behavior more pronounced when paired with ByteTrack, reducing utility in precise-count tasks.

The results emphasize that effective model-tracker pairing is critical, as sophisticated detectors alone do not guarantee accurate counts. For high-accuracy counting tasks, well-calibrated systems like YOLOv12 with BoT-SORT may outperform more complex alternatives.

These findings suggest broader applications where this perception model could support crop monitoring through integration with autonomous navigation platforms [14], though extension to different crop morphologies would require dedicated recalibration, as demonstrated by the specialized parameter optimization needed for citrus-specific detection systems [15].

Future work should explore deep learning-based tracking algorithms such as pseudo-labelling approaches [16] and enhanced association methods like StrongSORT [17] that could potentially offer improved accuracy and robustness compared to the traditional tracking methods evaluated in this study.

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